MIL-T-36974A (DSA-DM)
3 June 1975
SUPERSEDING
MIL-T-36974 (DSA-DM)
7 June 1973

MILITARY SPECIFICATION

TUBE, BLOOD COLLECTING, Vacuum, Non-Sterile

This limited coordination Military specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of MIL-T-36974 (DSA-DM). It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers non-sterile, vacuum, blood collecting tubes suitable for collecting, containing and transporting blood samples.

1.2 Classification. Non-sterile, vacuum blood collecting tubes shall be of the following types, sizes, colors and grades, as specified (see 6.1):

Type I - With anticoagulant
Type II - Without anticoagulant

Size 1 - 5 milliliters (ml.) nominal aspiration
Size 2 - 7 ml. nominal aspiration
Size 3 - 15 ml. nominal aspiration

Color codes

Red
Gray
Lavender
Blue

Grade A - Ammonium Oxalate and Potassium Oxalate as additive
Grade B - Lithium Oxalate as additive
Grade D - Sodium Fluoride as additive
Grade E - Disodium Edetate as additive
Grade F - Sodium Citrate as additive

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2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

DD-G-541	Glass, Laboratory.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.

Military

MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-66	Steel: Chemical Composition and Hardenability.
FED-STD-601	Rubber, Sampling and Testing.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

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AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary (N.F.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States (U.S.P.)

(Application for copies of the above publications should be addressed to the Mack Publishing Company, Easton, PA 18042.)

U.S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug and Cosmetic Act and Code of Federal Regulations
Title 21, Part 328

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20402.)

REAGENT CHEMICALS AND STANDARDS WITH METHODS OF TESTING AND ASSAYING By Joseph Rosin

(Application for copies should be addressed to D. Van Nostrand Company, Inc., Princeton, NJ 08540.)

3. REQUIREMENTS

3.1 Material.

* 3.1.1 Glass. Glass shall be in accordance with DD-G-541, Type I, Class B or Type II.

* 3.1.2 Rubber. The filler for rubber formulations shall contain at least 5% zinc oxide (by weight). The rubber shall be non-blooming, non-bleeding and self-sealing.

3.1.2.1 For Grades A through E. The rubber formulation shall contain at least 70% isobutylene isoprene polymer (by volume).

* 3.1.2.2 For Grade F. The rubber formulation shall contain at least 70% cis 1, 4 polyisoprene polymer (by volume) or shall be as specified in 3.1.2.1.

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* 3.2 Construction. The blood collecting tube shall consist of a tube and a stopper. The stopper shall fit snugly in the tube to form an airtight seal. Lubricants may be used on stoppers. Lubricants shall not adversely affect the results of the investigation for which the blood specimen is collected.

* 3.2.1 Tube. The tube shall be transparent, shall be fabricated of glass and shall be in accordance with Figure 2. Wall thickness shall be uniform. Bottom shall be round; open end shall be straight without a bead or a flare. Outside surface of the tube may have a label. If present, label shall contain blank space for use by hospital personnel.

* 3.2.2 Stopper. Stopper shall be fabricated of rubber and shall be in accordance with Figure 1 and applicable portions of Figure 2. Stopper shall be a diaphragm, molded in one piece. Top of diaphragm shall be flat with a recessed center, and bottom shall be a hollow plug. The lower portion of the stopper may contain recessed or raised integral rings or vertical slots. Coloring material shall be inorganic and non-bleeding. Shore durometer reading of stopper shall be 50 ± 5 when tested as specified in 4.4.1.

3.3 Color coding. Additives in tubes shall be differentiated by means of colored stoppers. Color of stopper and additive shall correspond as specified in Table I.

* Table I Stopper colors and additives

Stopper color	Grade	Additive
Red	None	None
Blue	A	6.0 milligrams (mg.) Ammonium Oxalate, U.S.P. and 4.0 mg. Potassium Oxalate, N.F.
Blue	F	0.5 ml. (± 0.05 ml.) of 3.8% aqueous solution of Sodium Citrate Reagent Grade (Rosin). Water shall be Purified Water, U.S.P.
Gray	B	15.0 to 22.5 mg. Lithium Oxalate, N.F.
Gray	D	27.0 to 34.5 mg. Sodium Fluoride, U.S.P.
Lavender	E	8 to 9.5 mg. Disodium Edetate Dihydrate

Note: Unless otherwise specified, tolerances shall be + 33-1/3%, -10%. Additives shall be powders except for Grade F. Powders shall be as specified in the applicable portions of the reagent section of the U.S.P. or the N.F.

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3.4 Performance.

3.4.1 Vacuum. When tested as specified in 4.4.2, minimum volumes of water drawn into tubes shall be as follows:

Size 1, Grade A--4.5 ml.
Size 1, Grade F--4.0 ml.
Size 2----------6.5 ml.
Size 3----------14.5 ml.

3.4.2 Resistance of tube to centrifugal force. Resistance of tube to centrifugal force shall be such that no breakage shall result when the tube is tested as specified in 4.4.3.

3.5 Date of delivery. Date of delivery shall be not more than 60 days after the date of manufacture.

* 3.6 Expiration date. The expiration date shall be not less than time indicated in Table II as measured from the date of manufacture. At the expiration date the blood collecting tube shall draw at least 90% of the volume specified in 3.4.1.

* Table II Expiration dates

Type and Grade	Expiration dates
Type I, Grades A, B, D and E	24 months
Type I, Grade F	18 months
Type II	36 months

* 3.7 FDA requirements. Blood collecting tubes shall be in accordance with HEW Food, Drug and Cosmetic Act and CFR, Title 21, Part 328, In Vitro Products for Human Use.

3.8 Workmanship. Workmanship shall be first class throughout. Tube shall be free of defects listed in Table V and other defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless

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disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the glass and rubber, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificates shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105. Inspection and acceptable quality levels shall be as indicated in Table III. Unit of product for examination purposes shall be the tube.

Table III For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	0.65
Minor defects	II	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, inspection level S-2 and Table IV.

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Table IV For tests

Unit of product	Characteristic	Requirement	Test procedure	AQL (percent defective)
Stopper	Durometer hardness	3.2.2	4.4.1	*
Tube	Vacuum	3.4.1	4.4.2	1.5
Tube	Resistance to centrifugal force	3.4.2	4.4.3	1.5

* Acceptance number shall be zero.

4.3 Examination. The tube shall be examined for defects including, but not limited to, those listed in Table V.

Table V Classification of defects

Categories	Defects
Major	
101	Tube not free of cracks or chips.
102	Stopper not secured in tube.
103	Tube not free of dirt or foreign matter.
104	Tube does not contain additive (except Type II).
105	Color of stopper not as specified.
Minor	
201	Tube not free of sharp edges.
202	Tube not free of bubbles.
203	Stopper not free of bloom.

4.3.1 Dimensional examination. The tube shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Durometer hardness. Stopper shall be tested in accordance with Method 3021 of FED-STD-601.

4.4.2 Vacuum. A seamless steel tube Brown and Sharpe Gage No. 20, 2-1/2 ± 1/16 inches long with a point at one end and a wall thickness from 0.0055 to 0.0060 inch or one of the needles listed below shall be used for testing (see 6.2). One end of the testing tube or the venipuncture end of the needle shall be placed

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in distilled water and the pointed end of the testing tube or the stopper end of the needle shall pierce the diaphragm.

* NSN's 6515-00-404-2240
6515-00-404-2260
6515-01-003-2368
6515-01-003-2369
6515-00-492-8712
6515-00-864-6788

4.4.3 Resistance to centrifugal force. Tube shall be filled to capacity with distilled water and suspended in a centrifuge which has an angle head of 50° from the vertical and a radius of 1-3/4 inches from the center of rotation to the center of the mouth of the tube shield. The centrifuge shall be accelerated gradually to a speed of 3000 revolutions per minute (r.p.m.). Centrifuge shall run at 3000 r.p.m. for 30 minutes. The current shall then be shut off and the centrifuge allowed to stop without the use of a brake.

* 4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table VI. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

* Table VI Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

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* 5.1.1 Unit of issue. One package, containing one hundred tubes as specified, constitutes one unit of issue.

5.1.1.1 Level A.

* 5.1.1.1.1 Unit package. Tubes (Grades A through E) shall be packaged in a fiberboard box of appropriate size and design. Box shall be designed with suitable interior fittings to insure against shifting of, or damage to, the contents. Closure shall be adequate to prevent accidental opening. As an alternate, unit package may be a fiberboard box with an open top shrink-wrapped in plastic. Box shall be designed with interior fittings to insure against shifting of, or damage to, the contents. Grade F tubes shall be packaged in two evacuated cans (50 tubes per can) and overpacked in a fiberboard box to comprise one unit. Closure shall be adequate to prevent accidental opening. When Grade F tubes are furnished with butyl rubber stoppers, tubes may be packaged as described above in lieu of the evacuated cans.

5.1.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shalll be Level A, B or C as specified (see 6.1).

* 5.2.1 Level A. Ten units of issue shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with PPP-B-636.

* 5.2.2 Level B. Ten units of issue shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Marking shall include the lot (control) number, contract or purchase order number, name of contractor, and the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59°-86°F.)". In addition, type II shelf life markings as specified in MIL-STD-129 shall be shown.

5.3.2 Exterior container. Each exterior container shall be marked in accordance with MIL-STD-129. Marking shall include the lot (control) number and the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59°-86°F.)". In addition, type II shelf life markings as specified in MIL-STD-129 shall be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Type, size, color and grade (if applicable) required (see 1.2).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 The items listed in 1.2 are to be used with the following items appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-404-2240	NEEDLE, HYPODERMIC, Blood Collecting, Double Pointed, 20 Gage, 1 inch, 12s
6515-00-404-2260	NEEDLE, HYPODERMIC, Blood Collecting, Double Pointed, 20 Gage, 1-1/2 inch, 12s
6515-00-492-8712	NEEDLE, HYPODERMIC, Blood Collecting, Double Pointed, Disposable, with shut-off sleeve, 20 Gage, 1/2 inch, 100s
6515-00-864-6788	NEEDLE, HYPODERMIC, Blood Collecting, Double Pointed, Disposable, with shut-off sleeve, 20 Gage, 1-1/2 inch, 100s
6515-01-003-2368	NEEDLE, HYPODERMIC, Multiple Sample, Double Pointed, Disposable, 20 Gage, 1-1/2 inch, with shut-off sleeve, 1000s

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National Stock No.	Item Identification
6515-01-003-2369	NEEDLE, HYPODERMIC, Blood Collecting, Double Pointed, Disposable, 20 Gage, 1-1/2 inch, 1000s
6630-00-404-2220	HOLDER, Blood Collecting Tube, 12s

* 6.3 Grade C of MIL-T-36974 (DSA-DM) has been deleted from this revision.

6.4 This specification covers the following items as listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6630-00-145-1132	TUBE, BLOOD COLLECTING, Vacuum, 5 ml., with Ammonium and Potassium Oxalate, 100s
6630-00-145-1137	TUBE, BLOOD COLLECTING, Vacuum, 7 ml., without Anticoagulant, 100s
6630-00-145-1143	TUBE, BLOOD COLLECTING, Vacuum, 15 ml., without Anticoagulant, 100s
6630-00-145-1527	TUBE, BLOOD COLLECTING, Vacuum, 15 ml., with Lithium Oxalate, 100s
6630-00-145-1533	TUBE, BLOOD COLLECTING, Vacuum, 7 ml., with Disodium Edetate, 100s
6630-00-145-1534	TUBE, BLOOD COLLECTING, Vacuum, 7 ml., with Sodium Fluoride, 100s
6630-00-250-4264	TUBE, BLOOD COLLECTING, Vacuum, 5 ml., with Sodium Citrate, 100s

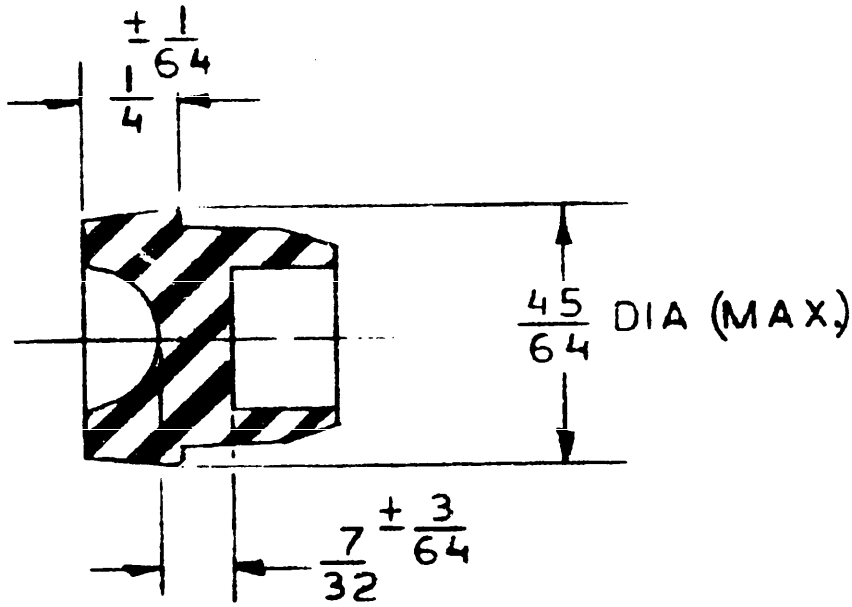
6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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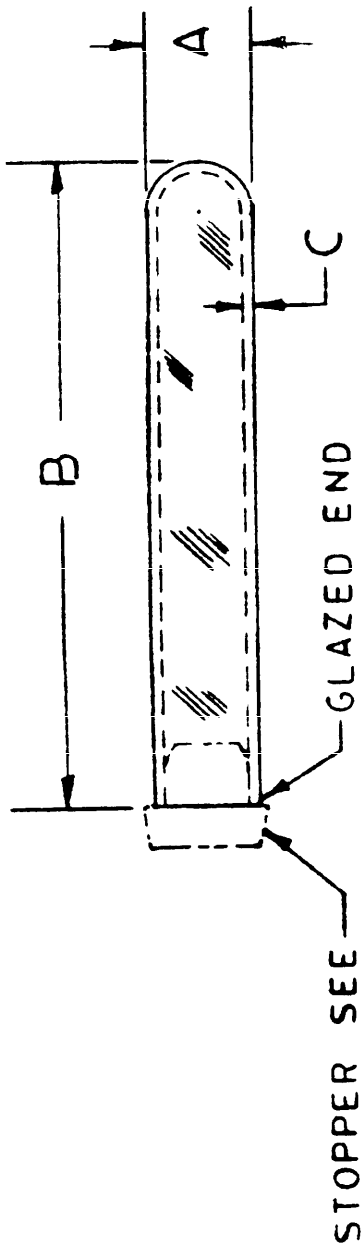
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DIMENSIONS ARE IN INCHES

FIGURE 1 - STOPPER

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SIZE	A	B	C
1 GRADE F	12-13	73-77	0.7 -1.1
EXCEPT 1&2 GRADE F	12-13	98-102	0.7 -1.1
3	15-16	125-129	0.8 -1.1

DIMENSIONS ARE IN MILLIMETERS

FIGURE 2-TUBE AND STOPPER

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-36974A (DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

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